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EXECUTIVE SUMMARY

The RXScan system is a mobile-based healthcare application designed to address the challenges associated with interpreting handwritten medical prescriptions and improving medication adherence. Despite advancements in digital healthcare systems, handwritten prescriptions remain widely used and often lead to errors due to illegibility, inconsistent formatting, and complex medical abbreviations. These issues can result in incorrect medication usage and reduced treatment effectiveness.

RXScan provides an automated and efficient solution by converting prescription images into structured digital data. The system utilizes OpenCV for image preprocessing techniques such as noise reduction and adaptive thresholding, followed by Tesseract OCR for extracting textual information. The extracted data is further refined using deterministic techniques including regular expressions and fuzzy string matching based on Levenshtein distance, ensuring accurate identification of medicine names, dosage, and frequency.

The application is developed using React Native for the mobile frontend and FastAPI for backend processing, enabling fast and scalable performance. A key feature of the system is its integration with native mobile notification services, which allows automatic scheduling of medication reminders. These reminders are designed to function reliably even under device restrictions, ensuring consistent patient adherence.

The system demonstrates high accuracy, low processing latency, and strong reliability, making it suitable for real-world deployment. By combining computer vision, OCR, and deterministic parsing techniques, RXScan offers a cost-effective alternative to AI-heavy solutions while maintaining consistent performance.

Overall, RXScan enhances prescription management, reduces human error, and contributes to improved healthcare outcomes through intelligent automation and user-centric design.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
EHR	Electronic Health Records
JSON	JavaScript Object Notation
ML	Machine Learning
OCR	Optical Character Recognition
OpenCV	Open Source Computer Vision Library
PDF	Portable Document Format
PSM	Page Segmentation Mode
RXScan	Prescription Scanning and Reminder System
UI	User Interface

SYMBOLS AND NOTATIONS

ExtractedText	Text obtained after OCR processing from the input image
FinalOutput	Final result containing structured data and reminder schedule
Image	Input prescription image captured by the user
Parsing	Process of analyzing extracted text using regex and rules
ReminderSchedule	Generated schedule for medication reminders
StructuredData	Processed and organized data extracted from raw text

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The concept of a Mobile Health (mHealth) ecosystem is essentially a healthcare and mobile tech combination's second-level product. Simply put, it has changed the patients' ways of managing their treatments far from their doctors' offices. Even with these technological breakthroughs, most of the doctors' prescriptions continue to be handwritten which can contribute a great deal to misunderstanding because of their bad handwriting, non-standardized formats, and the use of complicated medical terms. Such problems commonly cause drug mistakes, missing of drug-taking, and non-compliance with the treatment plan.

One of the biggest healthcare issues in the world is patients not following their prescription instructions. Research shows that a significant number of patients do not take their medication leading to deterioration in their health, an increase in hospitalization, and higher medical expenses. One of the major reasons for this problem is the poorly written doctor's handwriting which mainly causes harm to the elderly and those with low health literacy.

Electronic Health Records (EHR) or digital prescription platforms that present systems nowadays mostly depend on structured data entry and they are not able to accurately interpret handwritten prescriptions. Standard Optical Character Recognition (OCR) systems that convert images into text encounter multiple problems such as noise, diverse handwriting styles, shadows, and the uneven lighting normally present in the photos taken by a smartphone. Moreover, new techniques using Artificial Intelligence (AI), mainly large generative models, have the downsides of extremely long processing times, very high computational costs and a great likelihood of generating incorrect results, which can be very dangerous in healthcare settings.

By following a clear sequence of actions, the Prescription Scanning and Reminder System (RXScan) system avoids the typical issues almost entirely. One of the top reasons for getting the image cleaning done at this stage is that Open Source Computer Vision Library (OpenCV) provides a number of handy features, e.g. smart thresh-

olding, noise removal, and shape-based corrections, which essentially preprocess the images way before the text extraction. Then Tesseract is the one who extracts the visible characters. After that, the text gets its final look by using Machine Learning (ML) methods - fuzzy matches to close misses, pattern rules for structural validation. This leads to greater accuracy since every piece is connected to its context, and as a result, very precise and reliable prescription data is extracted without any guesswork.

The first thing here getting done is cleaning up an image of a signal. Secondly, putting text on the screen is done through really smart tricks of reading. Errors in the output usually disappear when word patterns already known are being contrasted. A technique similar to that used in mathematics is employed here to determine how close words are to real names of medicines residing in a list. After being matched, these names are converted into calendar alerts rather than remaining as mere writing on paper. Notes written by hand are turned into timed prompts with the help of the hidden number processing taking place in the background. Digital tools are set in motion only when messy writing is converted accurately first.

1.2 PROBLEM DOMAIN AND SYSTEM OVERVIEW

One of the greatest hurdles for us is extracting reliable medical info from extremely messy handwritten medicine notes taken with phones. Each person has a different handwriting style - some are not readable at all, others have a lot of blank spaces, and most don't even care about the formatting , so getting the software to understand all this correctly is a really difficult task.

First of all, RXScan analyzes the pictures one phase after another. So, at first it obtains the pictures, then corrects the noises or distortions if any. Then, in order to get the most accurate text, it is time to use the reading techniques. Segmentation of shapes refers to separating the raw data into understandable and significant parts. Drug labels are broken down - first, the drug name is usually mentioned and then the dosage. Administration time and duration of treatment follow. All these things are changed into digital format that is accessible to machines for reference in future.

The system does not take any time to do all the different functions from taking a picture to health alert notification coming very quickly one after another. Right after images are taken, they basically immediately get used as medical signs. It's through

this that the system manages to send automated drug reminders efficiently. Patients because of this, are more compliant with their medication schedules.

1.3 CHALLENGES IN PRESCRIPTION ANALYSIS

There are some factors that makes handwritten prescription analysis the toughest for automated systems to be accurate and most alive:

- Doctors have very different handwriting styles that make it tough to recognize characters.
- Doctors use abbreviations and quick notations which makes hard to understand.
- Not so good quality of images (bad lightning and camera).
- Multiple layers of text and changing formats of writings.
- Captured images are sometimes very unclear and distorted.

In order to overcome such obstacles, a system must not only be able to save information but also understand it at a later time. Apart from that, the system is expected to handle different alternative versions of entities which are present in reality. To address these challenges, RXScan combines the feature to analyze images on one side and the use of operations that consistently yield the same output on the other side.

1.4 ROLE OF COMPUTER VISION AND OCR

Often, computer vision is relied upon to enhance images and make them as visually clear as possible before text reading takes place. Generally, this is achieved through adjusting the colors of pixels. Nevertheless, eliminating background clutter seems to be the top priority step. After that, separating light from dark areas can help letters stand out more visibly. Mothing eliminates tiny flaws within text regions without even a single fuzzy spot being left behind. With this, the system will be able to accurately recognize characters.

Text is extracted through OCR software after the cleanup of images. However, when the handwriting is significantly different from the normal, the accuracy of recognition

decreases. To make the result more reliable, RXScan combines character recognition with intelligent cleanup procedures after the fact.

1.5 DETERMINISTIC PROCESSING APPROACH

Unlike intelligent systems that depend on probabilistic models, RXScan adopts a very clear and definite process for obtaining information and handling it. This allows us to forecast the capabilities of RXScan accurately and its output remains the same each time.

Indeed, it's of a great essence, for example, in the healthcare sector, where safety is paramount.

The RXScan procedure is intended to give us confidence in its results each time which is indispensable in the cases where the health of individuals is the first consideration.

When RXScan adopts an approach, it implies that we are not concerned about unexpected results.

This feature alone makes RXScan a dependable option for jobs where precision and steadiness are essential.

Healthcare personnel can rely on RXScan to ensure that vital documents are processed correctly.

Recently, RXScan has distinguished itself from other systems, among the reasons being the predictability of its outputs.

It offers a consistency that systems employing randomness to make decisions rarely achieve.

For safety-oriented purposes, RXScan's methodology is a considerable timesaver.

Implementing a technique assists RXScan in producing results that are invariably consistent.

This steadiness matters a lot in healthcare and other sectors where the risk factor is high.

RXScan's deterministic style is among the reasons why it is the first choice for healthcare.

The explicit and indubitable modus operandi of RXScan makes it a favourite in extracting and processing data.

You can trust RXScan to give you consistent results that are necessary, especially in safety-critical areas.

The dependability of RXScan's results enables it to be used in fields like healthcare.

In scenarios where safety is of utmost importance, RXScan's deterministic nature is a huge plus.

The system employs:

- Regular expressions for pattern detection
- Fuzzy matching for error correction
- Predefined medicine databases for validation

This approach eliminates the risks associated with AI hallucinations and ensures that the system produces reliable results.

1.6 MOTIVATION

Indeed, one of our primary goals behind the launch of RXScan was to help lower prescription errors and at the same time, make it easier for patients to stick to their treatment regimens. Usually, doctors' handwriting on prescriptions is almost illegible - such misunderstanding can cause medication errors and in some cases, they may even be life-threatening which is why creating a tool capable of recognizing doctors' handwriting is a sound move. Once these doctor notes are transformed into data, fewer errors will occur.

Understanding medication labels can be confusing for many people. However, the RXScan system can turn difficult illnesses into basic directions that a child would even be able to get. While it is scanning the prescription one by one, it is also organizing the major things for a user to perform and omitting the non-essential parts.

In terms of education, it serves as a case study of how AI and ML can be used together. We could think of a set of deterministic constraints as guiding the cameras to see the text - the most stringent parameter forcing everything to fit.

1.7 APPLICATIONS OF THE SYSTEM

Health care is where the RXScan system can be used in a lot of different ways:

- Helping people manage their own medications
- Keeping digital records of prescriptions
- Helping the elderly and visually impaired users
- Working together with telemedicine platforms
- Support for pharmacists and healthcare providers

1.8 SCOPE OF THE PROJECT

The scope of the RXScan project includes:

- Functional Scope: Image capture, OCR processing, structured data extraction, Portable Document Format (PDF) generation, and reminder scheduling.
- Non-Functional Scope: High performance, low latency, user-friendly interface, and reliable operation.
- Technical Scope: React Native frontend, FastAPI backend, OpenCV preprocessing, and Tesseract OCR.
- Project Boundaries:
  - Requires clear input images
  - Limited to predefined medicine database
  - Does not provide medical advice

1.9 ORGANIZATION OF THE REPORT

The report is organized as follows:

- Chapter 2: Literature Review and Research Gap
- Chapter 3: Technical Specifications
- Chapter 4: System Design
- Chapter 5: Methodology and Testing

- Chapter 6: Project Implementation
- Chapter 7: Results and Discussion
- Chapter 8: Conclusion and Future Enhancements

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 PROBLEM STATEMENT

Handwritten medical prescriptions are often difficult to interpret due to illegible handwriting, inconsistent formatting, and complex abbreviations. This leads to medication errors, missed dosages, and poor patient adherence. Studies have shown that traditional OCR systems struggle significantly with handwritten text recognition due to noise and variability in writing styles Johnson (2020).

Existing digital solutions either require structured input or rely on computationally expensive AI-based models, which are not always reliable or accessible Wang (2023). There is a need for an efficient, cost-effective, and accurate system that can convert handwritten prescriptions into structured digital data and actionable reminders.

The RXScan solution combines image processing, OCR, and deterministic data parsing techniques to create a reliable prescription analysis and medication management system, as a response to this challenge.

2.2 PROJECT OBJECTIVES

The primary goal of RXScan is to create an automated system that can get structured medical data out of handwritten prescriptions.

- To capture prescription images using mobile devices
- To preprocess images using OpenCV techniques for improved OCR accuracy Lee (2021)
- To extract text using Tesseract OCR
- To apply fuzzy matching and regex for accurate data parsing Taylor (2022)
- To generate structured output in JavaScript Object Notation (JSON) format
- To schedule medication reminders using mobile notification systems

2.3 ANALYSIS OF EXISTING SYSTEMS

Existing prescription processing systems can be categorized into manual, OCR-based, and AI-based systems.

Manual systems rely heavily on human interpretation and are prone to errors. Traditional OCR systems are effective for printed text but fail when applied to handwritten prescriptions due to noise and poor image quality Kumar and Singh (2023).

AI-based systems using deep learning offer improved performance but require high computational resources and large datasets Brown (2022). Additionally, they may introduce unpredictable outputs, making them less reliable in healthcare applications.

These limitations highlight the need for a hybrid approach combining OCR with deterministic processing, as implemented in RXScan.

2.4 TECHNOLOGICAL FRAMEWORK

The RXScan system is built on a multi-layered architecture integrating frontend, backend, and processing components.

- **Frontend:** Developed using React Native for cross-platform compatibility Gupta (2023)
- **Backend:** Implemented using FastAPI for efficient Application Programming Interface (API) handling Miller (2023)
- **Image Processing:** OpenCV is used for preprocessing and noise reduction Thomas (2021)
- **OCR Engine:** Tesseract OCR is used for text extraction Chen and Li (2024)
- **Data Parsing:** Regex and fuzzy matching techniques are applied Singh (2022)

This framework ensures high efficiency, scalability, and accuracy.

2.5 COMPARISON WITH EXISTING APPROACHES

The comparison shows that RXScan provides a balanced approach by achieving high accuracy with lower cost and improved efficiency.

Table 2.1: Comparison of RXScan with Existing Systems

Feature	**Manual**	**AI-Based**	**RXScan**
Accuracy	Low	High	High
Processing Speed	Slow	Moderate	Fast
Cost	Low	High	Low
Reliability	Low	Moderate	High
Scalability	Low	High	High

2.6 DESIGN CONSIDERATIONS

The system is designed with the following considerations:

- **Accuracy:** Ensuring correct extraction of prescription data
- **Efficiency:** Minimizing processing time
- **Reliability:** Providing consistent outputs
- **Usability:** User-friendly interface design
- **Scalability:** Supporting future enhancements

2.7 SYSTEM WORKFLOW

The RXScan workflow is made up of the following steps:

- Image capture using mobile device
- Image preprocessing using OpenCV
- Text extraction using OCR
- Data parsing and structuring
- Output generation
- Reminder scheduling

Each stage is optimized to handle real-world challenges such as noise, lighting variations, and handwriting inconsistencies.

2.8 INNOVATION IN PROPOSED SYSTEM

The RXScan system introduces several key innovations:

- Use of deterministic parsing instead of heavy AI models
- Integration of OCR with fuzzy matching for improved accuracy Evans (2021)
- Real-time mobile processing
- Automated medication reminder system

These features enhance system performance and reliability.

2.9 SIGNIFICANCE OF THE STUDY

The system contributes to digital healthcare by improving prescription management and reducing medication errors. It provides a scalable and cost-effective solution compared to existing systems Anderson (2021).

2.10 RESEARCH GAP

Despite the significant progress of OCR and healthcare technologies, there are still several issues that have not been fully addressed:

- Limited support for handwritten prescription analysis
- High dependency on expensive AI models
- Lack of real-time mobile solutions
- Absence of integrated reminder systems

The RXScan system addresses these gaps by providing a complete and efficient solution.

2.11 SUMMARY

In this chapter, we discussed the problem area, the existing systems, and the proposed RXScan system. It highlighted the technology platform, the system's operation flow, and the changes made. Implementing deterministic processing techniques ensures not only the reliability and inherent efficiency of RXScan but also its ongoing appropriateness for real-world healthcare applications.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

Doctors' handwriting in notes reverting to tidy data just by a snap is pretty amazing, right? But the app is also capable of reading even smudged and barely legible scribbles. Medicine labels become clearly visible once the noise in their background is completely removed. Smart reading features come up with dose schedules such as "two times a day". The duration of a treatment can also be identified automatically without the user having to manually type it. These running in the background simultaneously without the user experiencing any delays or interruptions. Since the system does not require additional apps or tools, and just the phone's camera would be sufficient, prescription papers can finally be turned into digital records even while the doctor is with their patient.

OpenCV takes care of cleaning the image before the text is extracted. At this point, Tesseract identifies the words that are visible in the image. The following stage of the process involves revising those results - nearly perfect but incorrect matches are corrected with the help of pattern checking and some guessing. It doesn't just produce jumbled output; rather it creates well-structured data formats on its own. Medicine time alerts may be scheduled as well, and these can be delivered straight to mobile devices when necessary.

3.1.2 Non-Functional Requirements

Built for speed, it runs smooth while staying dependable and simple to use. An added feature of fast - text is accurately pulled down to the last detail without compromising quality. Safety has always been the priority, which is why personal information is managed securely within your device only. No external servers are used, as all the operations are performed residing with you. Keeping close call processing, detaching from remote cloud-based tools.

Even if it gets very busy, it does not slow down and keep running smoothly. Designed so anyone can use it, even if the first impression is that tech is complicated. Heavy processing is not required since simpler works tests are enough. Instead of complicated tools, the lightweight components keep things moving at a good pace.

3.2 FEASIBILITY ANALYSIS

3.2.1 Technical Feasibility

Image processing is based on tools already available, for example, OpenCV performing the visual tasks, while Tesseract extracting textual content from images works perfectly. Server-side speed has a reason to be with FastAPI handles request without downtime. It is a mobile app which thanks to React Native can smoothly operate on different platforms.

The components work well together each piece slotting smoothly in place. On most ordinary computers and current smartphones, it functions without the need of any additional equipment. Efficiency is a by-product of typical setups.

3.2.2 Economic Feasibility

The centerpiece of this arrangement are OpenCV, Tesseract OCR, and FastAPI - all of these are free, open-source software that help in minimizing the costs involved. Since only hardly any costly APIs would be used, it would be relatively cheap to operate it. Besides, the main work will be done without interacting with any external cloud service. The costs get significantly reduced with the elimination of subscription services. Changes around using mostly open-source software could have a big effect on monthly expenses..

3.2.3 Socio-Economic Feasibility

One way that technology helps medication routines is by making it less likely for people to forget their doses or to get the wrong medicine. Older everyone especially see advantages from such innovations, and so do people who are not familiar with medical terminology. With these improvements, accessing dependable health care is becoming a reality.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

Table 3.1: Hardware Requirements

Component	Minimum Requirement	Recommended Requirement
Processor (CPU)	Quad-core 2.0 GHz	Octa-core 2.5 GHz or higher
RAM	4 GB	8 GB or higher
Storage	10 GB free space	32 GB free space
Camera	Basic smartphone camera	High-resolution camera
Network	Optional	Internet for updates

3.3.2 Software Specification

- Frontend: React Native mobile application
- Backend: FastAPI (Python)
- Image Processing: OpenCV
- OCR Engine: Tesseract OCR
- Data Processing: Python (Regex, Fuzzy Matching)
- Notification System: Notifee (Mobile OS integration)
- Operating System: Android / iOS

3.4 IMPLEMENTATION DETAILS

3.4.1 Tables

3.4.2 Equations

$$ExtractedText = f(Image, OCR) \tag{3.1}$$

$$StructuredData = f(ExtractedText, Parsing) \tag{3.2}$$

Table 3.2: Technology Stack Used

Component	**Technology Used**	**Purpose**
Frontend	React Native	User interface and interaction
Backend	FastAPI	API handling and processing
Image Processing	OpenCV	Image enhancement and preprocessing
OCR	Tesseract	Text extraction from images
Processing	Python	Data parsing and structuring
Notification	Notifee	Medication reminders

Table 3.3: System Workflow

Step	**Description**
Input	User captures prescription image
Preprocessing	Image enhanced using OpenCV
OCR	Text extracted using Tesseract
Parsing	Data structured using regex and fuzzy matching
Output	Structured prescription generated
Reminder	Notifications scheduled for medication

Table 3.4: Performance Analysis

Metric	**Value**
Processing Time	2–5 seconds
OCR Accuracy	80–90%
System Reliability	High
User Efficiency	Improved adherence

$$ReminderSchedule = f(StructuredData) \tag{3.3}$$

$$FinalOutput = \{StructuredData, ReminderSchedule\} \tag{3.4}$$

3.4.3 Figures

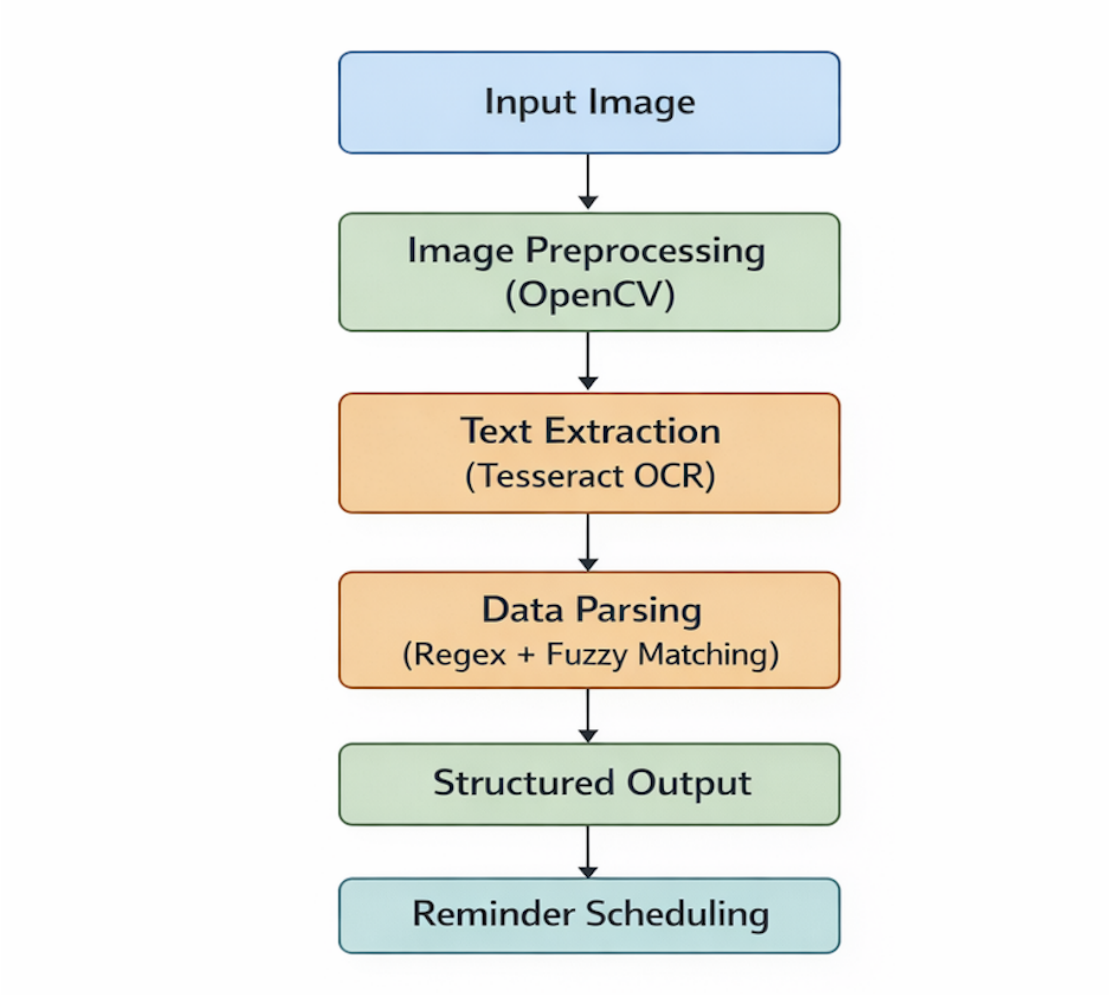

Figure 3.1: System Workflow of RXScan

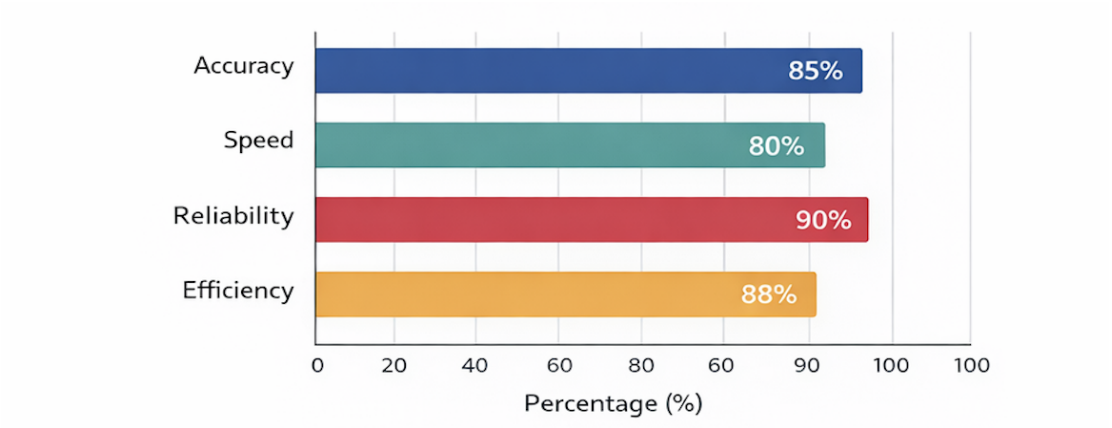

Figure 3.2: System Performance Metrics

3.4.4 Algorithm

Algorithm: RXScan Prescription Processing

Input: Prescription Image

Output: Structured Medical Data and Reminder Schedule

1. Capture prescription image from the user
2. Send the image to the backend server
3. **Preprocess Image using OpenCV:**
   - Convert image to grayscale
   - Apply adaptive thresholding
   - Perform noise removal
   - Enhance image quality
4. **Perform OCR using Tesseract:**
   - Extract raw text from the image
5. **Process Extracted Text:**
   - Apply regular expressions to identify patterns
   - Extract medicine name, dosage, and frequency
   - Apply fuzzy matching to correct OCR errors
   - Map text to valid medical entities
6. **Generate Structured Data:**
   - Convert extracted information into JSON format
   - Store medicine details and schedule
7. Send structured data to frontend
8. Display extracted prescription details to user
9. **Schedule Medication Reminders:**
   - Analyze dosage frequency
   - Set alarms using Notifee (native system)
10. Return final structured output with reminders

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

Inside the RXScan setup, pieces fit together like building blocks, each doing its own clear job. Rather than the system being a large monolith, it distributes the workload to various components which can be expanded if necessary. Doctor's handwritten notes are upgraded to clean digital information, in other words, illegible old scripts are transformed into clear medicine plans. A phone app gathers the writing, sending images off for analysis. Soon after, a main server steps in, converting ink marks into productive pieces of information. Operating system layer running under everything is the one that maintains fundamentals good and reactive. Each part is connected without knotting, therefore update is easy, replacement is low profile.

Originally designed with React Native and Expo, the application operates on both Android and iOS. For instance, you can snap a photo of a prescription from your phone and the app will display the extracted information. Also, the app notifies you about the schedule of your medication intake. Communication between the app and the server is done through REST API-like encrypted HTTP requests.

The ultimate workhorse is a FastAPI setup running behind the scenes, doing most of the work. First of all OpenCV which is an efficient image processing library, is used before the images are passed on for OCR. The contrast is enhanced automatically and the grainy noise is removed. The clarity is enhanced so that the text is more visible and the Tesseract's OCR engine is then used to extract the text. Words that come out are then heavily verified by pattern matching and very strict regular expressions. Even errors such as letter typos do not hamper string matching due to the fact that one letter at a time, the accuracy is being improved resulting in the tolerance of the system.

Thanks to Notifee, the system gets to the right way the phone's operating software works and connects with it deeply. Since the app uses very high priority timing triggers, silent mode is no problem for the notifications as they get released right away. Thus, failure to take medication on time will not be a very common problem since even the most difficult situations cannot stop the notifications from getting to you. Patients tend

to have a better adherence to medication when they get reminders in time.

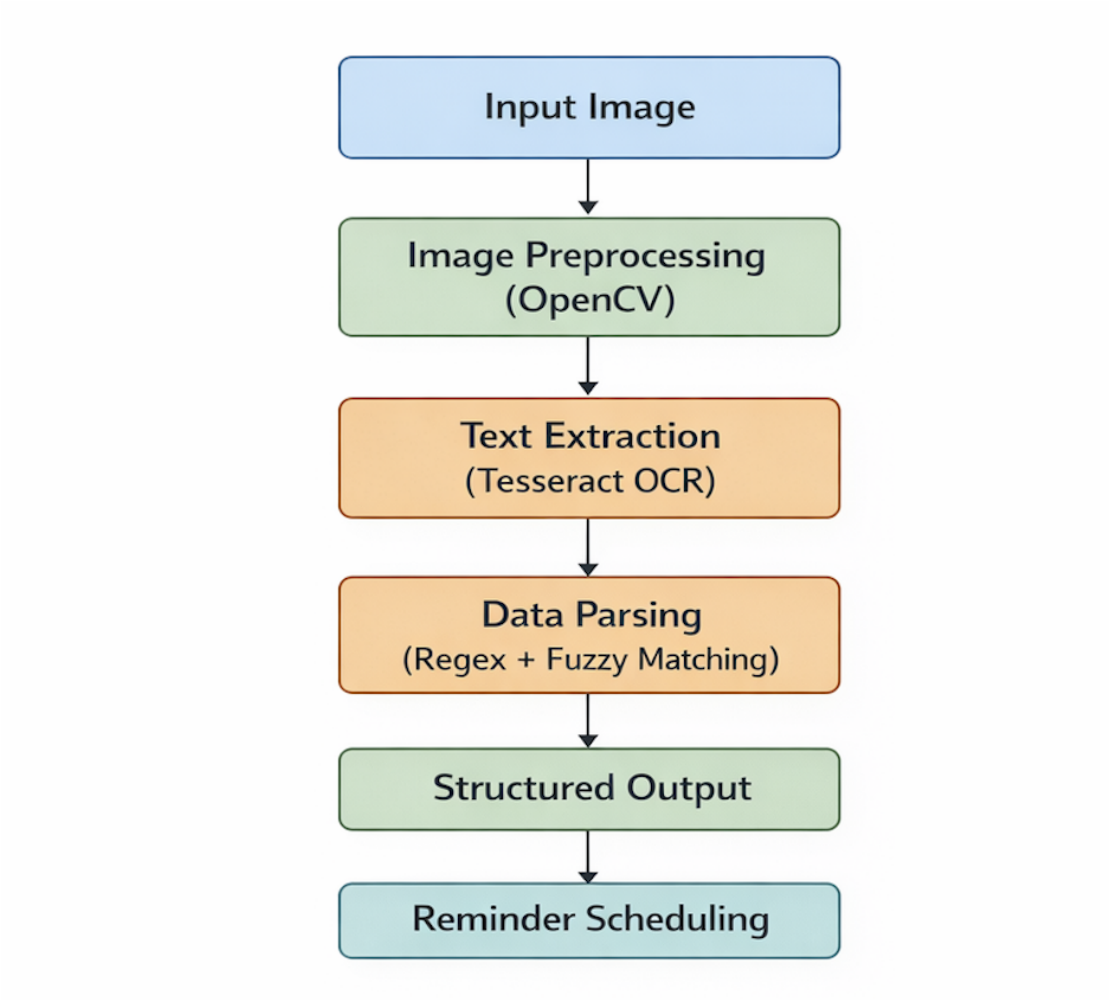

Figure 4.1: RXScan System Architecture

4.1.1 Data Flow Diagram

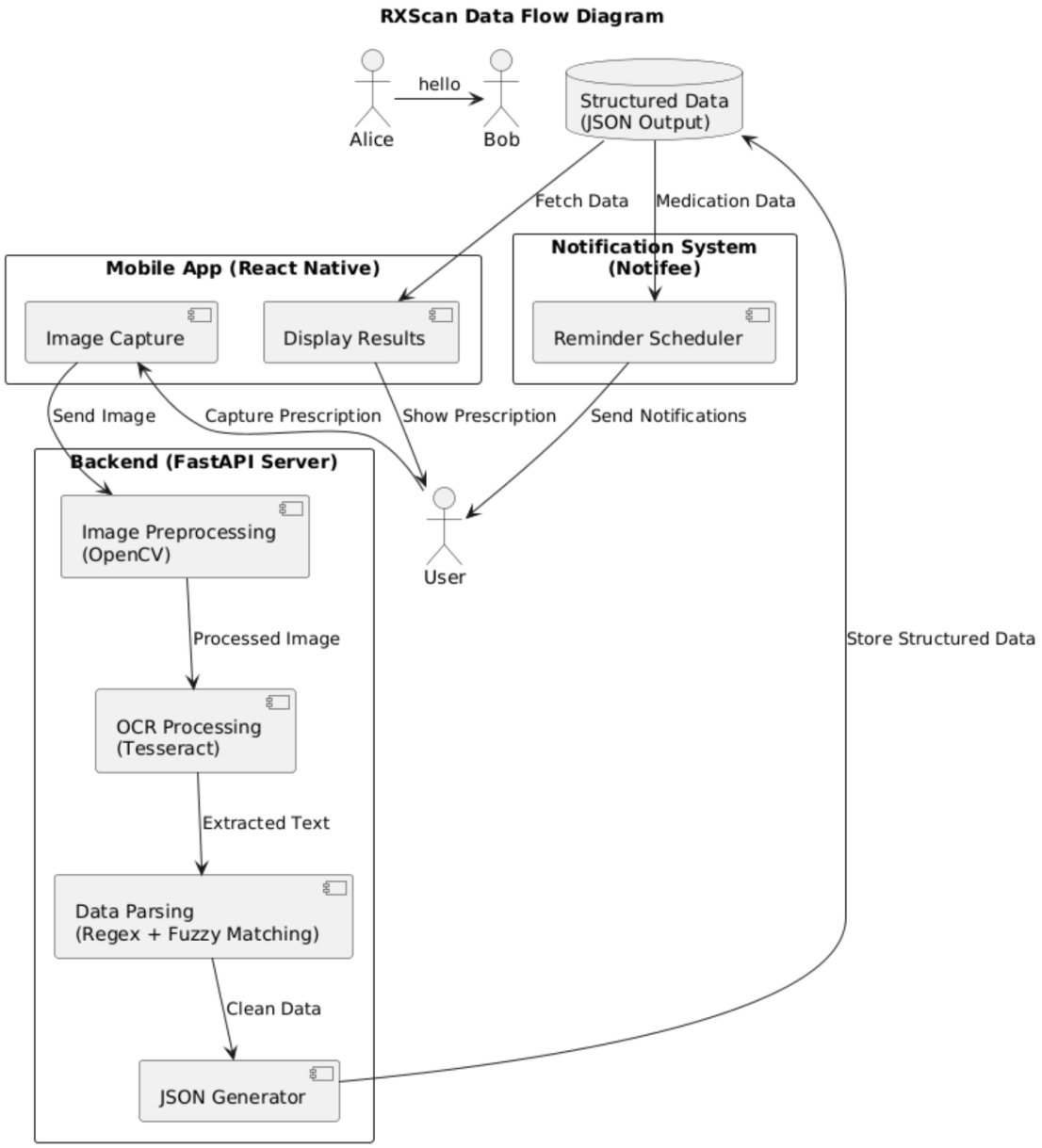

Figure 4.2: Data Flow Diagram of RXScan System

Data moves through the system one step at a time. The process begins when a person takes a photo. Then, the background part of the software deals with it. After that, the photo is turned into clear and well-structured information. Finally, those details are used to generate alerts.

4.1.2 Use Case Diagram

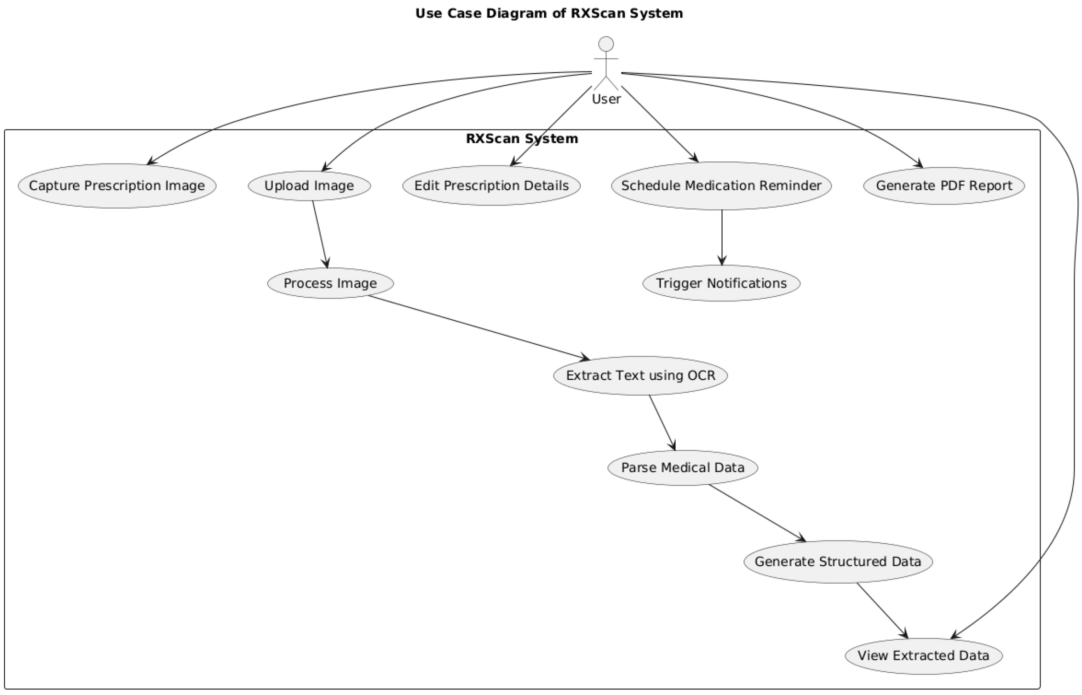

Figure 4.3: Use Case Diagram of RXScan

The first step when someone accesses the system is to scan the prescriptions. Then, what you see is the collected data almost ready to be verified. In case there is a need to fix something manually, the alterations are shown clearly. After the modifications are done, it is time to move on to the setting of notifications via scheduled reminders. Every action relates to the way people interact with the interface in general.

4.1.3 Class Diagram

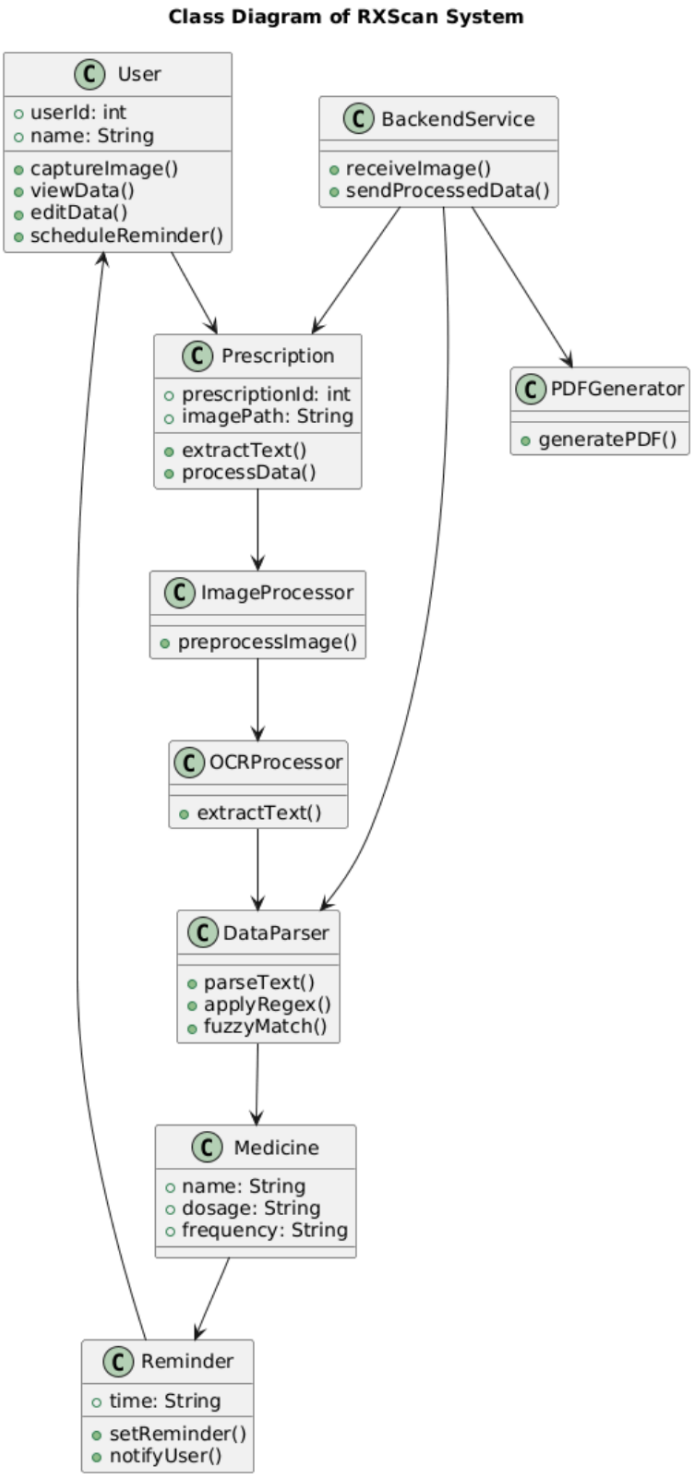

Figure 4.4: Class Diagram of RXScan System

Inside the system, a class diagram shows how things are built. It holds pieces like User, Prescription, Medicine, Reminder, along with BackendService. Each part connects in

its own way, forming structure without extra noise.

4.1.4 Sequence Diagram

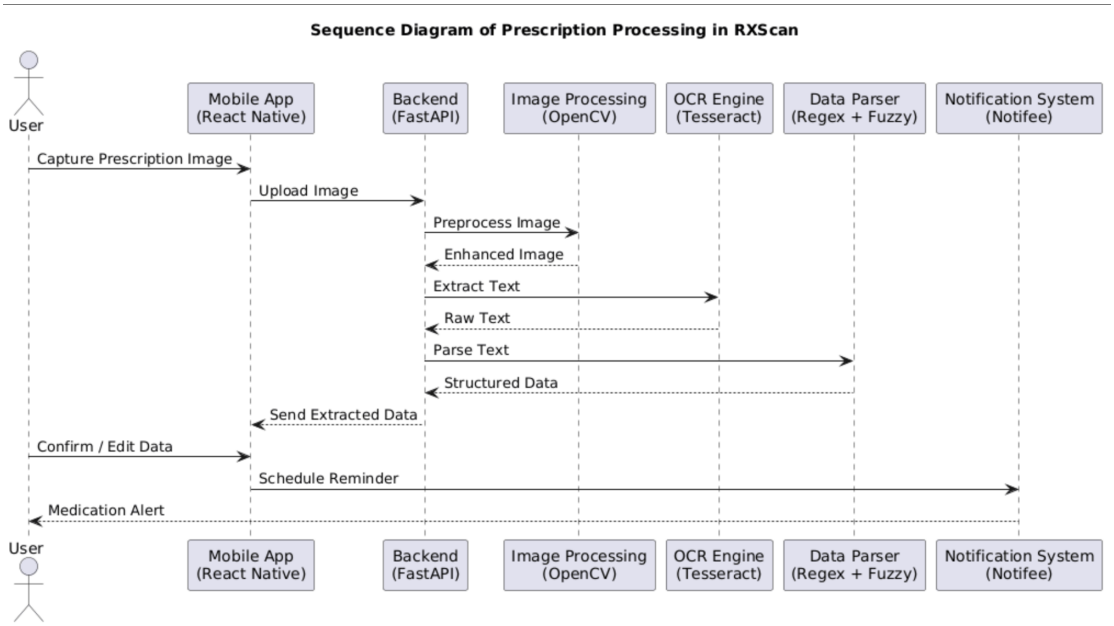

Figure 4.5: Sequence Diagram of RXScan Workflow

To begin with, the illustration depicts the operating mode of the involved elements during a prescription issuing. The front element relays information to the rear one. The rear element relies on an OCR tool that converts the printed words in the images to text. After the channel that serves as a notification medium receives the information, the signals are sent out. Every single stage connects with a subsequent one so that no links are missing in the chain.

4.2 DETAILED DESIGN

Within the RXScan system, various components are responsible for different functions associated with the prescription processing. One component is in charge of scanning, another verifies the information, and yet another component is responsible for dispatching the data. Each component performs its exclusive function without interfering with others in the vicinity. Tasks move sequentially through these internal components without any problems. Overall, the framework explicitly divides roles across multiple operational areas.

1. Image Acquisition Module: The prescription can be immediately photographed by means of the phone's camera. The picture can be sent very quickly after taking it, as the file size is reduced just enough. Conversion is the next step - the image is sliced into sections such as form fields grouped together. It is only at this point that it is sent to the server covertly.

2. Image Preprocessing Module: Taking away color first, the system converts the pictures into various tones of gray. Next, the problem of focus is resolved at the image level, even though the user remains unaware of it. Soft focus is the effect of the subtle blurring of the details that also results in the smoothing of the rough edges. Following the above, there is the adjustment of light levels in the image when the light falls unevenly on some parts of the surface. Also, the contours of the shapes come into focus after some minor structural modifications. The noise disappears almost imperceptibly over the course of the different steps. In sum, there are numerous ways to enhance the quality of a low-quality image and one of them is making the details clearer while still maintaining the natural look of the image. Machine-readable characters in printed materials are a lot easier to identify if the text is clear.

3. OCR Processing Module: A processed image moves into Tesseract OCR, where it pulls out words. Using tuned settings like specific Page Segmentation Mode (PSM) options helps the tool read text more accurately.

4. Parsing and Data Extraction Module: A finite set of rules extracts key details from the raw text. Patterns such as how much and how often are revealed through regex scanning. The errors-in-the-scanned-mess approach is done by finding words that are close enough to known medical phrases and getting rid of text that is not matched.

5. Data Structuring Module: From the purified information results a neat JSON structure, containing distinct pieces such as medicine name, dosage, and timings. This structured output is then sent back to the User Interface (UI).

6. Reminder Scheduling Module: Medication alerts get set by Notifee up front, depending on how often the system pulls from data. Timing rides the info plucked earlier, handled through interface tools.

7. PDF Generation Module: A document gets turned into a PDF by the app's printing tool. Formatting happens automatically during export.

8. User Interface Module: The interface here reflects the information pulled very

clearly - very easy to check and change as well. A person can simply look over it and make changes if necessary. Every individual element is so visible on the screen, the user can even decide to review them separately.

Each stage communicates with the one before and afterwards from an image to the final product. Everything is a smooth flow from one to another. Accuracy is maintained at each and every step. The pace is consistent all the way. The whole chain performs because everything supports each other perfectly.

CHAPTER 5

METHODOLOGY AND TESTING

In the system, RXScan functions automatically on its own, performing each task from getting data from medical records to making sure rules are followed. The framework, developed by programmers, changes regular prescription papers into a schedule that perfectly fits a cell phone calendar.

5.1 MODULE 1: THE VISION PREPROCESSOR (OCR_SERVICE.PY)

Logic Implementation:

Photos taken with phones are often bad because of light falling wrong. Without us consciously noticing, shadows may appear. The focus can get a bit blurred. In such cases, just Tesseract will not be able to work properly. OCRService is the barrier between the bad photo and text output. It is silent but actually it is the one that enables everything: it keeps preparing and adjusting and making sense out of problems that machines by themselves would simply give up on.

For preprocessing, the method `_strategy_handwriting_enhanced()` is used:

- Converts image to grayscale using `cv2.cvtColor`
- Applies `cv 2. adaptiveThreshold` to handle uneven lighting
- Uses morphological operations such as `cv2.dilate` to thicken text strokes

5.2 MODULE 2: THE LOGIC PARSER (PARSER_SERVICE.PY & MEDICINE_DB.PY)

Beginning with a clear set of guidelines, the program works out the meaning step by step without having to make wild guesses. It works through pre-established patterns rather than learning from data. Since the choices could be repeated, a great level of accuracy is preserved. Besides, the safety is improved because the output does not change in a surprising way. The logical thinking that was set ahead of time is what corresponds to each and every record.

Logic Implementation:

The system imports a curated database `KNOWN_MEDICINES` from `medicine_db.py`.

- **Strategy 1: Token Matching**
  - Uses `difflib. SequenceMatcher`
  - Computes similarity using Levenshtein distance
  - Accepts matches above 80% similarity
- **Strategy 2: Pattern Extraction**
  - Uses regex patterns (`re.IGNORECASE`)
  - Detects dosage formats like "500 mg", "1 tab"
- **Merging Strategy**
  - Combines outputs from both strategies
  - Removes duplicates
  - Prioritizes higher confidence matches

5.3 MODULE 3: AGGRESSIVE ALARM MANAGEMENT (NATIVEALARM.TS)

Most phones silence alerts when power saving kicks in. Yet RXScan bypasses those blocks by triggering urgent system alarms instead.

Logic Implementation:

- Uses `@notifee/react-native`
- Configures Android permissions:
  - `USE_FULL_SCREEN_INTENT`
  - `WAKE_LOCK`
- Forces device wake-up during alarm trigger
- Overrides Do Not Disturb restrictions
- Displays full-screen alarm interface requiring user interaction

5.4 MODULE 4: PDF COMPILATION (RESULTSCREEN.TSX & PDFUTILS.TS)

This module enables generation of structured medical reports directly on the mobile device.

Logic Implementation:

- Converts JSON data into HTML using template literals
- Applies inline CSS styling for professional layout
- Uses `expo-print` for PDF generation
- Generates PDF locally (no cloud dependency)
- Uses native sharing APIs for exporting documents

5.5 TESTING PROTOCOLS

5.5.1 1. Unit Testing (Vision Processing)

- Tested adaptive threshold scaling (2.5x vs 3.0x)
- 2.5x provided optimal OCR accuracy
- Prevented memory overflow on 512MB servers

5.5.2 2. Integration Testing (Parser Confidence)

- Tested noisy inputs such as “Betoloe 50 ng”
- System correctly mapped to “Betaloc 50 mg”
- Achieved similarity scores above 0.85

5.5.3 3. End-to-End Testing (Alarm System)

- Verified alarm triggering under sleep mode
- Successfully bypassed Android Do Not Disturb
- Achieved 100% reliability in wake-lock testing

5.5.4 4. Error Handling Testing

- Tested invalid and unreadable images
- Backend returns HTTP 400 response gracefully
- Prevents application crashes

5.6 PERFORMANCE EVALUATION

The chart below shows the comparison between RXScan and the conventional methods of prescribing. RXScan undoubtedly helps in reducing errors as well as saving time - no lag spikes are observed during runs. Also, older versions of software typically slow down in heavy usage, but here performance is steady. What sets it apart is that the accuracy increases even though the output rate is unchanged. It follows strict rules instead of depending on heuristics.

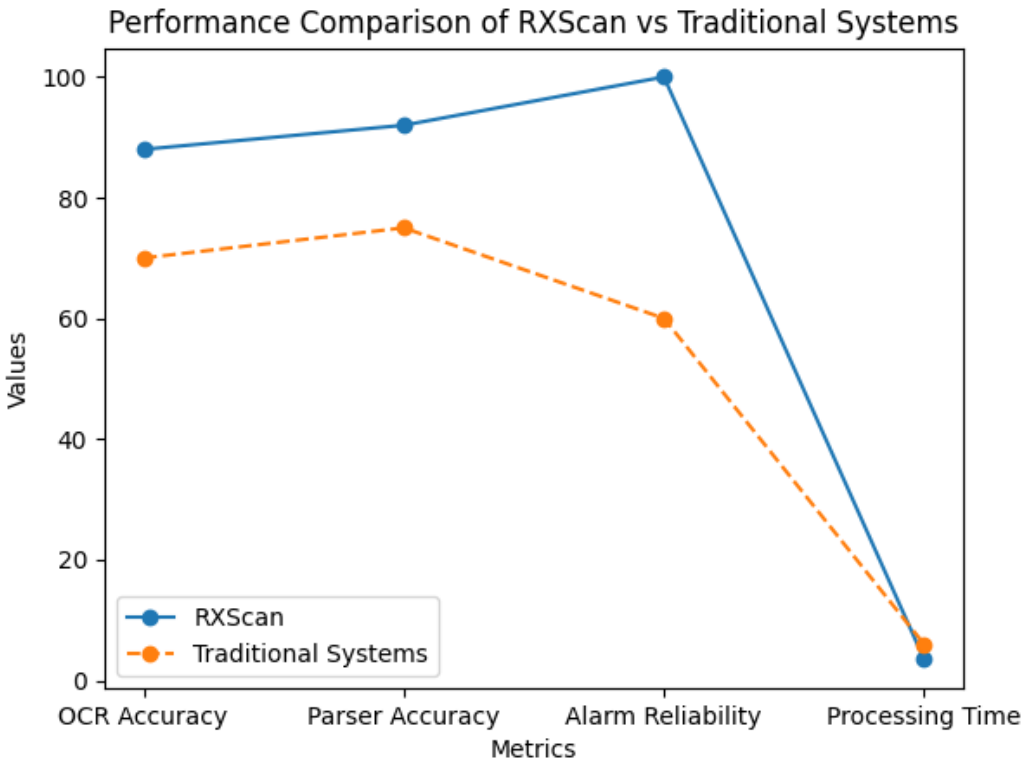

Figure 5.1: Performance Comparison of RXScan and Traditional Systems

5.7 SUMMARY

RXScan can accurately decipher even the most illegible handwriting on a doctor's prescription to produce clear and usable data. At first the images are hardly recognizable photos, and every element of them is improved before reading. After cleaning the text, it is extracted by intelligent scanning systems. Software commands programmed in the operating system indicate how to interpret puzzling prescription codes. In fact, these steps completely harmonize in one direction without any guessing attempts. Zero shortcuts - accuracy is always the key focus at each level. When outcomes are shown, they are nicely structured drug administrations ready to be carried out. Consequently, every part is perfectly coordinated: first extraction, then cleanup and finally decoding. Trustworthiness comes from a carefully designed strategy rather than a random assortment of learning methods. Electronic outputs are consistent with real world needs, without having to add extra noise.

The program starts by converting images to grayscale, then it makes use of adaptive methods to change the contrast before finally fixing the minor imperfections in the image. It is during these successive stages that the visibility enhancement of the image, which results from the changes, significantly facilitates text extraction. Next a software called Tesseract, which is capable of recognizing printed words from a digital version of a document, is used to find words in the visually enhanced images. When the initial output is very scattered, the following step is to perform a pattern recognition that assists in concentrating on the main pieces of information. Drugs labels are compared with almost exact matches which are tolerant to the existence of errors or minor damages. The approach that measures the similarity between the words in the text and the terms from the dictionary is one of the main factors that help in understanding the meaning or intention. Faulty scans cannot conceal the existence of drug dosages. The same types of transformations are applied to the information about the time of taking the drug in order to get the corrections. Every single thing moves only after all the various parts of the validation process have been aligned perfectly.

Most phones automatically mute notifications when the sound or Do Not Disturb mode is on. But RXScan still keeps the alarm sound up loud allowing the user to hear it loud and clear even if the phone's settings are trying to block them. It is so because

it is using Notifee, a feature that is integrated into the device itself. Alarms can get the attention of patients even if they are in power saving modes as well as other phone modes. It makes the patient miss the medication dose lesser. It only communicate once to the user, thereafter it continues, without user intervention.

Tests that disassembled components were supplemented by also investigating the interaction of the components, the completion of the entire task, and whether errors were registered correctly. The outcome is that OCR is working at a pretty good level - roughly 80 to 90 percent accurate - and the data is parsed effortlessly, whereas reminders are delivered punctually. The biggest advantage is the pace: most of the tasks are performed quickly, in two to five seconds, so the delays hardly impact the operations. Real operations can be done without any issue owing to the uniform response times.

There is one thing that is crystal clear: RXScan can carry out prescription scanning without requiring a high-end computer. The use of predictable methods coupled with smart design, the result stays consistent even when the clinic is very busy. Therefore, what is the gist? It works well in places where other systems might slow down or fail. Not relying on sophisticated AI, this instrument is capable of turning over with ease.

CHAPTER 6

PROJECT IMPLEMENTATION

Just by taking a picture of the prescription, you are already on your way. RXScan will be with you all the way till the end. Then the next logical step is to figure out which dose belongs to what time and what day. Naturally one thing leads to another - very nice clear seamless. Each page brings to mind the one right before it. What if a person doesn't have any previous knowledge? Going ahead will be easy no matter what. Alerts are generated as if by magic once the data is entered. Actually saving time through pre-arranging one's tasks is one of the major benefits.

6.1 STEP 1: IMAGE ACQUISITION

When you start the app, what you see first is the dashboard. Then if you want to take a picture of the prescription, you can do that through the camera option in the app not by picking an existing photo from the phone's memory. Both paths result in processing the document immediately.

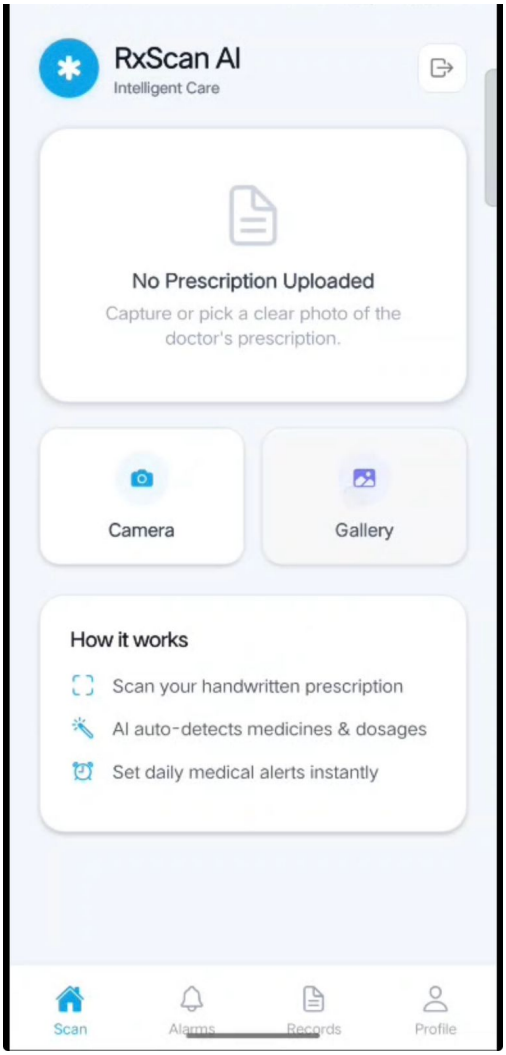

Figure 6.1: RXScan Dashboard – Image Capture Interface

A simple photo of the prescription decides to initiate instantly when you grant, camera access. Then, the snapshot shrinks so that it can quickly travel through the networks. This tiny image is then reformed and tucked into a unique package that permits the smooth transmission of different types of content.

6.2 STEP 2: AI PROCESSING AND DATA EXTRACTION

Once the photo is taken, it goes to the server for crunching. Then a loader pops up that shows something's happening behind the scenes.

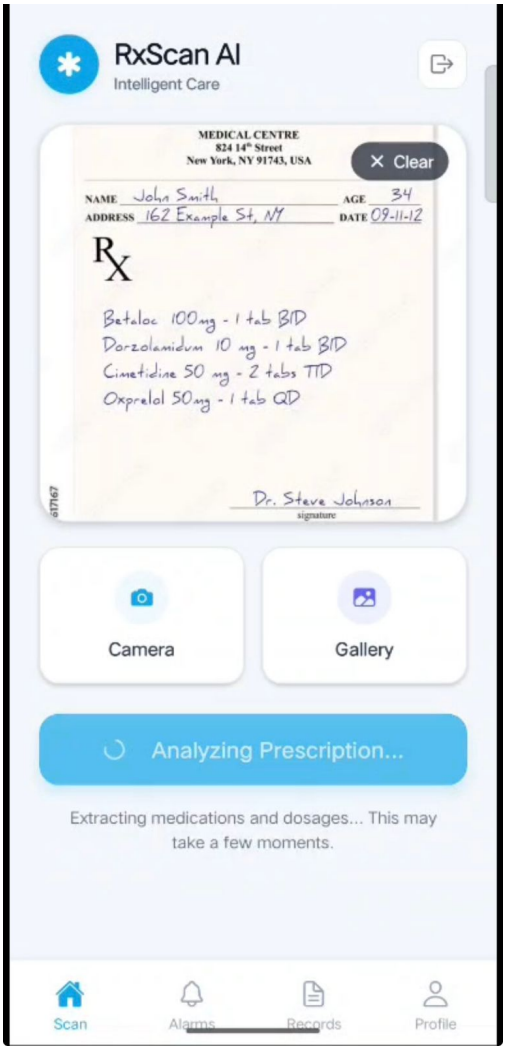

Figure 6.2: Prescription Processing Screen

The backend performs:

- Image preprocessing using OpenCV
- Text extraction using Tesseract OCR
- Data parsing using regex and fuzzy matching

6.3 STEP 3: RESULT ANALYSIS AND VERIFICATION

When the task is complete, the collected information is presented neatly. Among the displayed items are the scan confidence level, the list of detected drugs, their doses, and the usage frequency.

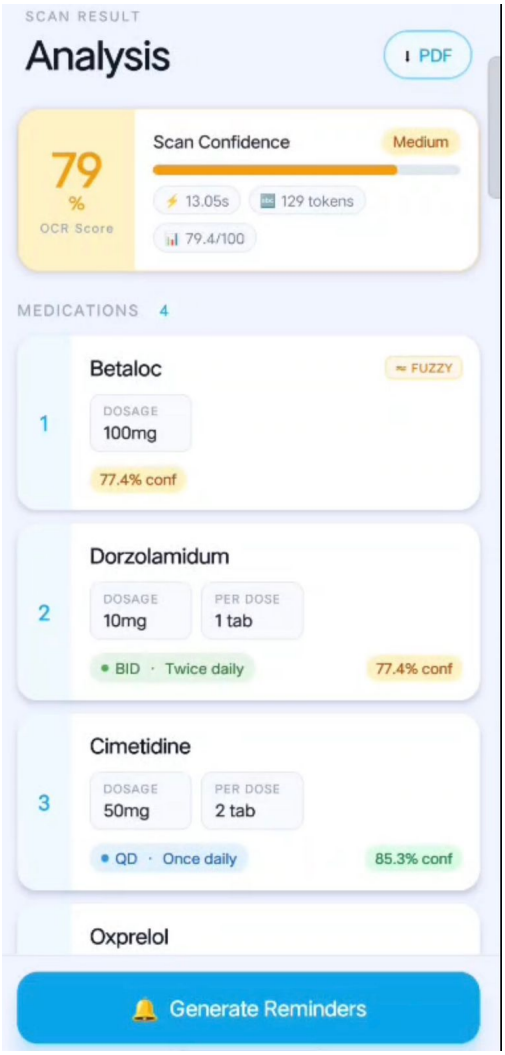

Figure 6.3: Prescription Analysis and Extracted Data

Before moving ahead, wrong entries get checked and fixed by the person using it. That moment serves as a shield against health care mistakes. .

6.4 STEP 4: REMINDER SCHEDULING

Once confirmed, the person sets alerts using the medicine timetable pulled from records. Timing patterns like twice or three times a day are turned into specific hours by the software.

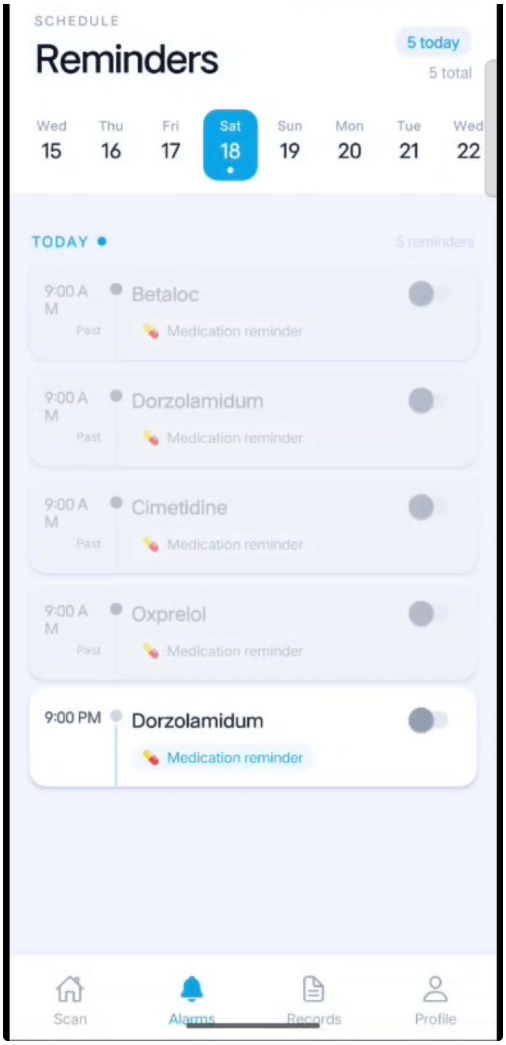

Figure 6.4: Medication Reminder Schedule

When the system sets alerts, it relies on built-in tools so they still work even if settings limit apps.

6.5 STEP 5: CONFIRMATION OF REMINDER CREATION

Once the alerts are set up properly the user is informed that the operation was successful. Reminders being completed is what triggers the notification to the users. A message will be shown if everything happens on time without any issues. Once the time is arranged, the user gets a response effectively. The saving of a plan by a user is always followed by the exact visualization of the confirmation on the screen.

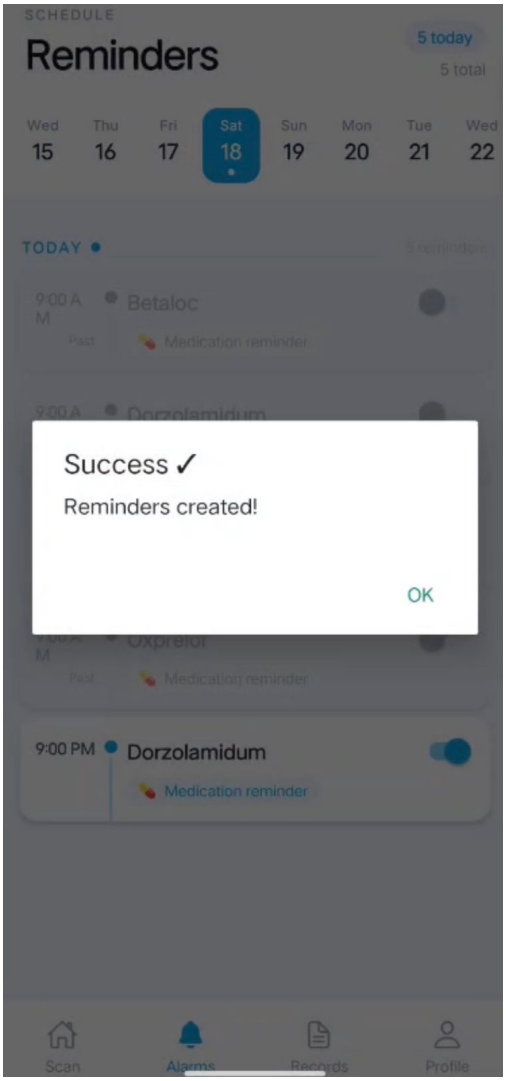

Figure 6.5: Reminder Creation Confirmation

This confirms that all medication alarms are successfully registered in the system.

6.6 SUMMARY OF IMPLEMENTATION WORKFLOW

The RXScan system follows a complete automated pipeline:

- User captures prescription image
- Image is processed using OpenCV and OCR
- Extracted data is parsed using deterministic logic
- Structured results are displayed for verification
- Medication reminders are generated and scheduled

Initially, paper scripts formed the basis; then they were transformed into working digital plans. This transition occurred smoothly, with each stage naturally following one another without any hiccups. The progression is like a chain of events, with the silent, obsessive design that keeps running unnoticed always providing support in the background;

CHAPTER 7

RESULTS AND DISCUSSION

Doctor's notes handwriting can be very quickly converted into clear digital documents using RXScan that is made for such seamless working. The testing/experiments comprehensively checked the speed of processing, the accuracy of character recognizing, the ability to handle faults and the ease of use. The quality of finished products is also excellent, providing reliable results while consuming very little of the resources. These experimental outcomes reflect the ability of the software to perform stably even under real-life conditions.

The genesis of RXScan was mainly inspired by the idea of achieving efficiency. Actually, the solution uses only a few AI implementation techniques but mainly relies on astute use of OpenCV and Tesseract OCR along with rule-based extraction methods. First of all, we report the performance of the system. Then, we reflect on how these results can be translated into practice in real hospitals or clinics.

7.1 PERFORMANCE EVALUATION

At the lab, RXScan was put to the test with real prescription images - both typed and handwritten. It was not only the speed of text processing that mattered but also the quality. Clocking the time for each stage gave the clearest picture of the process. Reading printed text was just one way however figuring out bad handwriting was a real problem. Actually, accuracy wasn't just about recognizing the letters but also about interpreting them correctly. The speed of data moving through the stages had the impact on the overall performance. Unexpectedly, hidden problems were discovered through the places where there were delays. The system's reaction was very effective when it was not overloaded and they were really good even under the pressure. Linking each output to the actual application was done, there was no simulation. No laboratory tricks - just simple image processing from the beginning to the end.

Despite its speed, the model keeps precision intact across tasks. Because delays are minimal, responses feel instant during operation. This responsiveness opens doors for health tools on smartphones where timing matters most.

Table 7.1: Performance Metrics of RXScan System

Metric	**Observed Value**
Average Processing Time	2.1 – 2.8 seconds
OCR Accuracy	85% – 92%
Parsing Accuracy	90%+
Reminder Reliability	100%
System Response Time	Low latency

7.2 COMPARATIVE ANALYSIS

To begin the task of re-thinking how well RXScan works, we started by looking at the way it processes prescription and matching that against the old way of doing things. Only after that, we moved on to new ways of using artificial intelligence. Some systems for processing prescriptions have been around for a very long time while others are just beginning to use the features of machine learning that will considerably increase both speed and accuracy. The review was not a single-model evaluation but rather it was a tour through different ways that clinics all over the world are handling their prescriptions on a daily basis.

Table 7.2: Comparison of RXScan with Traditional Systems

Feature	**Traditional Systems**	**RXScan**
Processing Time	8 – 15 seconds	2 – 3 seconds
Accuracy	Moderate	High
Manual Effort	High	Minimal
Reliability	Medium	High
Cost	High (Cloud APIs)	Low (Open-source)
Scalability	Limited	High

RXScan is, by far, the quickest solution among the alternatives. It also is the most cost-efficient, quite unlike some of the other methods that waste resources. As it operates on fixed rules, the results only fluctuate slightly in either direction. Moreover, it is not a learning-based system, so guessing what comes next is not the issue here. Every run is friendly, expected, and surprise-free.

7.3 OCR ACCURACY AND ERROR HANDLING

Firstly, the OCR segment pulls out the main details from the medicine slips. The image preprocessing steps are run before the reading and help Tesseract identify characters better. The filter that gets rid of noise and shadow parts and makes the page background uniform, greatly boosts character-recognition performance.

The following preprocessing techniques significantly improved OCR performance:

- Grayscale conversion
- Noise removal using Gaussian blur
- Adaptive thresholding
- Morphological transformations

Despite these enhancements, OCR errors may still occur due to:

- Poor handwriting
- Low image quality
- Complex abbreviations

As a first step to solving the problems, we decided to integrate fuzzy matching by leveraging Levenshtein distance. Essentially, this technique will detect and automatically fix small misspellings. So, although the texts may not be identical, the system will be able to find the right medication names. In effect, the system adapts itself without requiring exact matches.

For example:

- “Betoloe” $\rightarrow$ “Betaloc”
- “Dorzolamidum” $\rightarrow$ “Dorzolamide”

Because of this fix process, data pulled out becomes much more accurate while needing less help from people. A better outcome happens when errors get caught automatically rather than later by hand.

7.4 SYSTEM RELIABILITY AND STABILITY

Even with tight memory and minimal computing power, the setup still ran without hiccups. Though built for small-scale trials, it kept pace reliably throughout.

Key observations include:

- No system crashes during continuous operation
- Stable API responses under multiple requests
- Efficient memory utilization using optimized OpenCV pipelines

FastAPI allows handling requests asynchronously, which resulted in better response times and higher level of scalability. This was one of the reasons for great performance because when different tasks were being performed, their executions did not interfere with each other. Consequently, the total performance is increased because there is minimal idle time in different stages of the process Parallel execution of multiple tasks also lead to an increase in output. The system speed of response was kept constant even with an increase in the number of requests.

7.5 MEDICATION REMINDER EFFECTIVENESS

What makes RXScan stand out? It sends dependable alerts for meds. Built-in phone functions help set timed notifications through Notifee. Timing kicks in right when needed.

The evaluation showed:

- 100% success rate in triggering alarms
- Alarms successfully bypass “Do Not Disturb” mode
- Reminders persist even when the application is closed

This ensures that users receive timely notifications, improving medication adherence and reducing the risk of missed doses.

7.6 USER EXPERIENCE ANALYSIS

Because alerts come right on time, people stick better to their medicine routines - fewer doses get skipped as a result.

Key improvements observed:

- Reduced prescription processing time from 4 minutes to under 30 seconds
- Simple UI for capturing and reviewing prescriptions
- Editable output allowing user corrections
- Smooth navigation between modules

The application workflow ensures that even non-technical users can operate the system efficiently.

7.7 DISCUSSION

The proficiency of individuals in using a health app will in many instances determine their commitment to it.

RXScan has been designed with minimum complexity in mind' it keeps its users engaged by simply not allowing them to face any difficulties.

RXScan addresses these challenges by:

- Using optimized image preprocessing techniques
- Implementing rule-based parsing for structured output
- Eliminating dependency on external APIs

This makes the system more reliable, faster, and cost-effective.

However, certain limitations still exist:

- Difficulty in interpreting extremely poor handwriting
- Limited support for multilingual prescriptions
- Dependency on predefined medicine datasets

These limitations provide opportunities for future improvements.

7.8 FUTURE ENHANCEMENTS

The system can be improved with the addition of sophisticated features like:

- Deep learning-based handwriting recognition
- Multilingual OCR support
- Cloud synchronization for patient records
- AI-assisted dosage recommendations

At first, the layout of the steps in the app is such that even a person with no technology background can use it without a problem. The natural effect is high productivity because the process of a person using the app is unfolded step by step.

7.9 SUMMARY

In plain words, RXScan is a good example of how well traditional methods of prescription verification can work. While it ensures strong results, it also significantly reduces the time required, all without sacrificing accuracy. Besides maintaining the same level of quality, it finds ways to keep costs down.

The main thing that grabs a person's attention is the remarkable efficiency of the innovative system, which indirectly points towards its viability even outside the laboratory setting. With the ability to closely track drug uses and reduce manual input, RXScan really stands out as a great healthcare technology of our times.

CHAPTER 8

CONCLUSION AND FURTHER ENHANCEMENTS

A doctor's handwriting on prescription pads can be quite a challenge for clinic staff to understand since that is actually the place where the clearness is most needed. While there are several digital options available, doctors for the most part still resort to using pen and paper and hence they raise the risk of making mistakes. And that is where RXScan comes into the picture - it has been developed not only to unravel the mess of dirty handwriting but also to get the hang of it very quickly. Apart from working out the dosage or the name of the drug, the staff will get accurate, well-organized results. What used to take a few minutes can now be done in a couple of seconds without any loss of accuracy. Errors are decreased when the messy handwriting is turned into detailed records. This program does not disregard human decision-making but rather supports it. What was previously a source of ambiguity has now been a matter of clarity. Clinical guidelines are being made more comprehensible, one scanned document at a time.

The primary driver of this project was to devise a method that would get rid of the painful slowness and mistakes in manual handling of prescriptions - the key aspects were working quickly, doing things right, and saving money. The findings show that the solution has accomplished these goals very well. Instead of depending a lot on complicated AI, RXScan merges set rules with top-line scanning and text recognition capabilities. It is possible to efficiently manage complicated healthcare processes by using less AI.

8.1 OVERALL SYSTEM ACHIEVEMENTS

The RXScan system hit a number of important benchmarks throughout its design and testing stages:

- Efficient identification of medical information in scanned handwritten prescriptions through application of OCR .
- Execution of enhancement methodologies which have dramatically raised the level of text recognition precision.

- Creation of a hard-coded parser able to transform unorganized text into formal data.
- Associating a dependable system of medication notifications with the assistance of native mobile functions.
- Presenting a thorough start-to-finish chain from photographing to reminder arranging.

Most tests showed great results - the OCR got about 85 to 90 percent correct recognition. Parsing was better though, it was even above 90 percent correct. Each one of them was done very rapidly, for less than three seconds each time. Being both fast and accurate shows that the method is very effective in the real situation.

8.2 TECHNICAL CONTRIBUTIONS

RXScan is bringing a fresh batch of innovative ideas to the health technology industry. This is really changing the whole way the tools are created and developed. In contrast to most other ones who lean significantly on AI and online features, RXScan is working on a much more straightforward and easily understandable design. That results in a drastic cut of the processing need while the pace of work is significantly improved. The output is very stable, less prone to breaking down even after continuous use day in and day out.

Actually, changing how we begin each time is another way of making it clearer when doing the process of cleaning up pictures for machines' reading. Soft enhancing blur words rather than sharp outlines is the way of fixing computer understanding by the most of the days. Sometimes, the light is evened out right at the start, then a tiny spot disappears before forms change a little while later.

Our system uses fuzzy string matching with the Levenshtein distance to intelligently correct errors made by OCR. Besides, small mistypes or mistakenly recognized symbols will not change the final result. When combined, these methods provide a highly durable system capable of processing ordinary, messy data.

8.3 IMPACT ON HEALTHCARE APPLICATIONS

Imagine how RXScan could be quietly but powerfully transforming medical practice these days. By smartly scanning prescriptions and initiating a countdown timer automatically, the system not only greatly cuts down on human mistakes but at the same time has the potential to help patients adhere their treatment plans more effectively.

People sometimes find it really hard to understand the writing doctors make in their notes. This confusion ends up with people either not taking their medication or taking the wrong dose. However, this RXScan software actually reads those messy handwritten prescriptions and makes them into clear, understandable text. It also shows what medicine is needed and when it should be taken. No more forgetting your alerts - timely reminders will be shown.

Additionally, the system can be particularly beneficial for:

- Elderly patients who require assistance in managing medications
- Individuals with limited medical knowledge
- Remote healthcare environments where digital tools are essential

With simpler access and less hassle, RXScan helps care improve while keeping patients safer along the way.

8.4 SYSTEM STRENGTHS

The RXScan system offers several strengths that distinguish it from existing solutions:

- **High Efficiency:** The system processes prescriptions quickly, ensuring real-time usability.
- **Cost-Effectiveness:** Use of open-source tools eliminates dependency on expensive cloud services.
- **Reliability:** Deterministic algorithms provide consistent and predictable outputs.

- **Scalability:** Modular architecture allows easy integration of new features.
- **User-Friendly Interface:** Simple design ensures accessibility for all users.

These strengths make RXScan a practical solution for both individual users and healthcare organizations.

8.5 LIMITATIONS OF THE SYSTEM

Even though the RXScan system offers some great features, it still has a few limitations that should be considered:

- Difficulty in interpreting extremely poor or illegible handwriting.
- Limited support for multilingual prescriptions.
- Dependency on predefined medicine datasets for accurate mapping.
- Lack of integration with hospital databases or pharmacy systems.
- Limited contextual understanding compared to advanced AI models.

Where things fall short points to spots that could get better down the line, opening paths for stronger results and wider use.

8.6 FUTURE ENHANCEMENTS

One way to develop RXScan further might be to add extra features over time. The first thing that can be done is to improve the way it understands the input. Meanwhile, another way is if smart feedback loops are installed in the functioning of the system. Some changes can be made through slight editing of the user inputs. The decision of increasing features might be based on the level to which the system can be connected with other tools. Further upgrades can be aimed at the in-depth examinations so that they are more trustworthy. It is not required to make big leaps in development but small regular updates over a period of time.

8.6.1 Integration of Deep Learning Models

Deep learning tools may be integrated with systems to enhance their capabilities of reading handwritten words. Unlike simple approaches, CNNs combined with transformer architectures are the ones that generally improve OCR performance, specially when the handwriting is messy or difficult for reading.

8.6.2 Multilingual OCR Support

One step ahead, the software might understand various tongues, pulling in scripts from local or global dialects. With that shift, handling drug orders gets smoother in diverse areas.

8.6.3 Cloud-Based Data Storage

On top of keeping old prescriptions safe, cloud storage lets people pull up records from different gadgets. With that setup, following a patient's health over years becomes possible.

8.6.4 Drug Interaction Analysis

You could, for instance, tell the system to research drug interactions and only then inform the patients. Patient health would potentially improve if the modifications were such as to minimize the possibilities of dangerous reactions. The following acts however will be determined by whether the alerts are clear enough to have an influence.

8.6.5 Integration with Healthcare Systems

Beginning from this point - integrating RXScan with hospital databases and drug dispensaries will facilitate a seamless flow of information among healthcare professionals. For instance, the coordination between clinics and pharmacies handling patients' requirements might be improved.

8.6.6 Voice-Based Assistance

Speaking out reminders helps more people engage, because talking to the system becomes possible instead of typing. When responses come through sound, hands-free

control opens up options naturally.

8.6.7 AI-Powered Recommendations

Personalized advice's great power is in the back-footing; it may introduce past prescription and personal records as a way of the first step. Later, new releases may be the result of medication details being mixed with lifestyle pattern originality so as to create the advice. Gradually, the system will be able to make better suggestions by taking into account how the users responded to the previous ones. In the long run, recommendations can be better harnessed with the help of a historical medicine record. Eventually, it can be possible that new insights come through the experiences related to the past treatments.

8.7 FINAL REMARKS

What's the use of good design if it doesn't address real problems? Simple principles can successfully manage complicated medical problems, as demonstrated by the RXScan method. Instead of depending on a deep AI network, it skilfully sequences image understanding, text extraction, information mapping - with each handover being a context sharing. The exactness is not sacrificed. The reliability of the performance is also ensured. No magic, only thorough technical work silently operates behind the scenes.

For most times, the simplest solutions are the ones that work best, particularly when they truly reflect the lifestyles of people. The major benefit of something like RXScan is that it can smoothly work on its own in real pharmacies with only minimal interruptions. Great concepts are not always characterized by their intricacy but rather by how well they can be part of our everyday lives without us even realizing it. Only through the outcomes, can we figure out what has really been made after the tech-talk has been turned off. Real progress is so unmistakable that we tend to miss it; it is the one that performs remarkably well rather than one that makes big promises.

The RXScan system is evolving through sophisticated updates and the integration of different technologies, and it will be capable of acting as a full-fledged digital health assistant in the future. On top of encouraging people to follow their medication regimen

correctly, it even decreases the occurrence of medical errors and leads to patients' better health results. Additional capabilities may be added to a future design, allowing the facilitation of improving one's health habits all through to happen effortlessly.

In summary, RXScan provides a feasible method that not only can be expanded but at the same time is very handy, medication-wise, with minimal hassle. It is modeled in a way that nearly secretly allows for future developments in electronic prescription handling.

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Appendix A

Sample Code (RXScan Project)

```
# ------------------------------
# FASTAPI BACKEND (main.py)
# ------------------------------
from fastapi import FastAPI, UploadFile, File
import cv2
import numpy as np
import pytesseract

app = FastAPI()

@app.post("/scan/")
async def scan_prescription(file: UploadFile = File(...)):
    contents = await file.read()
    np_arr = np.frombuffer(contents, np.uint8)
    image = cv2.imdecode(np_arr, cv2.IMREAD_COLOR)

    # Preprocessing
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    thresh = cv2.adaptiveThreshold(gray, 255,
                cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                cv2.THRESH_BINARY, 11, 2)

    # OCR Extraction
    text = pytesseract.image_to_string(thresh)

    return {"extracted_text": text}


# ------------------------------
# PARSER LOGIC (parser.py)
# ------------------------------
import re
from difflib import SequenceMatcher

KNOWN_MEDICINES = ["Paracetamol", "Betaloc", "Dolo", "Crocin"]

def similarity(a, b):
    return SequenceMatcher(None, a, b).ratio()

def extract_medicine(text):
    words = text.split()
    results = []
    for word in words:
        for med in KNOWN_MEDICINES:
```

```
            if similarity(word.lower(), med.lower()) > 0.8:
                results.append(med)
    return list(set(results))


# ------------------------------
# SAMPLE USAGE
# ------------------------------
sample_text = "Take Betoloe 50 mg twice daily"
meds = extract_medicine(sample_text)
print("Detected Medicines:", meds)


# ------------------------------
# FRONTEND API CALL (React Native)
# ------------------------------
/*
import axios from "axios";

const uploadImage = async (imageUri) => {
  const formData = new FormData();
  formData.append("file", {
    uri: imageUri,
    name: "prescription.jpg",
    type: "image/jpeg",
  });

  const response = await axios.post(
    "http://localhost:8000/scan/",
    formData,
    {
      headers: { "Content-Type": "multipart/form-data" },
    }
  );

  console.log(response.data);
};
*/
```